

REPLACING RELATIVE PROJECTIVITY BY PROJECTIVITY FOR THE TRACE-CLASS CONVOLUTION MODULE

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ABSTRACT. We answer Remark 5.14 of Crann’s paper *Amenability and covariant injectivity of locally compact quantum groups II*. For the trace-class convolution modules in Theorems 5.12 and 5.13, relative 1-projectivity may be replaced by ordinary metric 1-projectivity. More precisely, for every locally compact quantum group $\mathbb{G}$, compactness of $\widehat{\mathbb{G}}$, relative 1-projectivity of $(\mathcal{T}(L^2(\mathbb{G})), \triangleright)$ as a left module over itself, and 1-projectivity of the same module are equivalent. The analogous right-module statement for $(\mathcal{T}(L^2(\mathbb{G})), \triangleleft)$ also holds.

1. SOURCE QUESTION

The source paper proves the following theorem, in its notation [1, Theorems 5.12 and 5.13]. For a locally compact quantum group $\mathbb{G}$, the conditions

$$\widehat{\mathbb{G}} \text{ compact,} \quad \mathcal{T}(L^2(\mathbb{G})) \text{ relatively 1-projective in } (\mathcal{T}(L^2(\mathbb{G})), \triangleright) - \mathbf{mod}$$

are equivalent; there is also a right-module analogue for $(\mathcal{T}(L^2(\mathbb{G})), \triangleleft)$. Immediately after those theorems, Remark 5.14 asks whether the word “relative” can be removed from condition (2) [1, Remark 5.14].

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2. PROOF INTUITION

There are two different reasons why the replacement works.

First, ordinary 1-projectivity is strong enough to recover the relative splitting. Apply projectivity to the multiplication

$$m = (\Gamma^r)_* : \mathcal{T}(L^2(\mathbb{G})) \hat{\otimes} \mathcal{T}(L^2(\mathbb{G})) \longrightarrow \mathcal{T}(L^2(\mathbb{G})).$$

This gives exact splittings whose norms are $< 1 + \varepsilon$. Dualizing puts us inside the dual operator space of normal completely bounded maps

$$\mathcal{B}(L^2(\mathbb{G}) \otimes L^2(\mathbb{G})) \longrightarrow \mathcal{B}(L^2(\mathbb{G})).$$

A weak-star cluster point as $\varepsilon \rightarrow 0$ has norm at most 1 and is still a module left inverse of Γ^r . Passing back to the predual gives the norm-one relative projective splitting.

Second, the reverse implication uses the special trace-class convolution algebra rather than plain operator-space projectivity. If $\widehat{\mathbb{G}}$ is compact, then $\mathbb{G}$ is discrete. Hu, Neufang, and Ruan show that in this case $(\mathcal{T}(L^2(\mathbb{G})), \triangleright)$ has a right identity obtained by extending the identity of $L^1(\mathbb{G})$. A completely contractive Banach algebra with a contractive right identity is automatically 1-projective as a left module over itself.

3. THE RESULT

Theorem 1. *Let $\mathbb{G}$ be a locally compact quantum group and set*

$$A_r = (\mathcal{T}(L^2(\mathbb{G})), \triangleright).$$

The following are equivalent.

- (1) $\widehat{\mathbb{G}}$ is compact, equivalently $\mathbb{G}$ is discrete.
- (2) A_r is relatively 1-projective in $A_r - \mathbf{mod}$.
- (3) A_r is 1-projective in $A_r - \mathbf{mod}$.

Theorem 2. *Let*

$$A_l = (\mathcal{T}(L^2(\mathbb{G})), \triangleleft).$$

The following are equivalent.

- (1) $\widehat{\mathbb{G}}$ is compact, equivalently $\mathbb{G}$ is discrete.
- (2) A_l is relatively 1-projective in $\mathbf{mod} - A_l$.
- (3) A_l is 1-projective in $\mathbf{mod} - A_l$.

Lemma 1. *Let A be a completely contractive Banach algebra.*

- (1) *If A has a right identity e with $\|e\| \leq 1$, then A is 1-projective in $A - \mathbf{mod}$.*
- (2) *If A has a left identity e with $\|e\| \leq 1$, then A is 1-projective in $\mathbf{mod} - A$.*

Proof. We prove the left-module statement; the right-module statement is identical with sides reversed. Let $q : Y \rightarrow Z$ be a complete quotient morphism in $A - \mathbf{mod}$, and let $\phi : A \rightarrow Z$ be a completely bounded A -module morphism. Given $\varepsilon > 0$, choose $y \in Y$ such that

$$q(y) = \phi(e), \quad \|y\| < \|\phi(e)\| + \varepsilon \leq \|\phi\|_{\text{cb}} + \varepsilon.$$

Define $\widetilde{\phi}(a) = a \cdot y$. The module action is completely contractive, so $\|\widetilde{\phi}\|_{\text{cb}} \leq \|y\|$. Moreover

$$q\widetilde{\phi}(a) = a \cdot q(y) = a \cdot \phi(e) = \phi(ae) = \phi(a),$$

since e is a right identity. Thus A satisfies the metric lifting property with constant 1. $\square$

Lemma 2. *Let $A_r = (\mathcal{T}(L^2(\mathbb{G})), \triangleright)$. If A_r is 1-projective in $A_r - \mathbf{mod}$, then A_r is relatively 1-projective in $A_r - \mathbf{mod}$.*

Proof. Let $H = L^2(\mathbb{G})$. The right fundamental unitary gives a normal complete isometry

$$\Gamma^r : \mathcal{B}(H) \rightarrow \mathcal{B}(H) \overline{\otimes} \mathcal{B}(H),$$

and the multiplication $m = (\Gamma^r)_* : A_r \widehat{\otimes} A_r \rightarrow A_r$ is a complete quotient morphism in $A_r - \mathbf{mod}$. Since A_r is 1-projective, for each $n \geq 1$ there is an A_r -module morphism

$$\sigma_n : A_r \rightarrow A_r \widehat{\otimes} A_r$$

such that $m\sigma_n = \text{id}_{A_r}$ and $\|\sigma_n\|_{\text{cb}} < 1 + 1/n$.

Dualizing, $S_n = \sigma_n^*$ is a normal completely bounded module map

$$S_n : \mathcal{B}(H) \overline{\otimes} \mathcal{B}(H) \rightarrow \mathcal{B}(H)$$

with $S_n \Gamma^r = \text{id}_{\mathcal{B}(H)}$ and $\|S_n\|_{\text{cb}} < 1 + 1/n$. The space of normal completely bounded maps $\mathcal{CB}^\sigma(\mathcal{B}(H) \overline{\otimes} \mathcal{B}(H), \mathcal{B}(H))$ is a dual operator space. Hence a subnet of (S_n) has a weak-star cluster point S of cb-norm at most 1.

The equations defining $S_n \Gamma^r = \text{id}$ and the module-intertwining identities are weak-star closed: after pairing with elements of the predual A_r , they are linear equations. Therefore $S \Gamma^r = \text{id}_{\mathcal{B}(H)}$, and S is still the corresponding normal module morphism. Since S is normal, it has a preadjoint

$$\sigma = S_* : A_r \rightarrow A_r \widehat{\otimes} A_r.$$

The relations above give $m\sigma = \text{id}_{A_r}$, $\|\sigma\|_{\text{cb}} \leq 1$, and σ is an A_r -module morphism. The density identity

$$A_r = \overline{\text{span}}(A_r \triangleright A_r)$$

is proved in the source paper, so this is exactly relative 1-projectivity in the non-unitized form used there. $\square$

Proof of Theorem 1. The equivalence of (1) and (2) is Theorem 5.12 of the source paper [1, Theorem 5.12].

If (3) holds, Lemma 2 gives (2), and hence (1).

Conversely assume (1). Then $\mathbb{G}$ is discrete. Hu, Neufang, and Ruan show that if f_0 is the identity of $L^1(\mathbb{G})$ and $\omega_0 \in \mathcal{T}(L^2(\mathbb{G}))$ is a normal extension of f_0 , then ω_0 is a right identity for $(\mathcal{T}(L^2(\mathbb{G})), \triangleright)$ [2, Proposition 5.1(iii)]. The discrete counit f_0 is a normal state, and the normal Hahn–Banach extension theorem gives a norm-one normal extension ω_0 to $\mathcal{B}(L^2(\mathbb{G}))$. Lemma 1 therefore gives 1-projectivity of A_r in $A_r - \mathbf{mod}$. Thus (1) implies (3). $\square$

Proof of Theorem 2. The source paper’s Theorem 5.13 gives (1) $\Leftrightarrow$ relative 1-projectivity of A_l in $\mathbf{mod} - A_l$ [1, Theorem 5.13]. The proof of Lemma 2 applies verbatim to the left fundamental unitary and the multiplication $\triangleleft$, with right modules in place of left modules, giving ordinary 1-projectivity $\Rightarrow$ relative 1-projectivity.

For the reverse implication, assume $\mathbb{G}$ is discrete. The unitary antipode induces a complete isometric anti-isomorphism $R_* : A_r \rightarrow A_l$ satisfying

$$R_*(\omega \triangleright \tau) = R_*(\tau) \triangleleft R_*(\omega).$$

If ω_0 is the contractive right identity of A_r , then $R_*(\omega_0)$ is a contractive left identity of A_l . Lemma 1 gives 1-projectivity in $\mathbf{mod} - A_l$. $\square$

4. VERIFICATION AND NOVELTY NOTES

The argument is formal after two external inputs:

- (1) Crann’s Theorems 5.12 and 5.13 [1], which identify relative 1-projectivity of the two trace-class modules with compactness of $\widehat{\mathbb{G}}$.
- (2) Hu-Neufang-Ruan’s Proposition 5.1(iii) [2], which says that when $\mathbb{G}$ is discrete, a normal extension of the identity of $L^1(\mathbb{G})$ is a right identity for the trace-class convolution algebra $(\mathcal{T}(L^2(\mathbb{G})), \triangleright)$.

The local run indexes were searched for the arXiv id, the title, “relative 1-projectivity”, “TCQ”, and trace-class projectivity; no duplicate settlement was found. Web and local source searches found the source paper, the HNR trace-class convolution paper, Crann’s 2021 local-theory paper, and later trace-class convolution papers, but no explicit answer to Remark 5.14.

5. REFERENCES

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